



[BIT703 Web Technologies](#) / [Contents](#) / [Topic 12: Web application performance](#) / [Types of tests](#)

Nonfunctional testing 60m



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[Skip to main content](#)

Nonfunctional testing can be performed at any individual test level or all test levels, preferably as early as possible to avoid uncovering significant nonfunctional defects in a system or product that is about to be released.

Nonfunctional testing is designed to evaluate how the product behaves according to the nonfunctional specifications. Sometimes this is known as 'expectations' testing because every customer will expect the product to be accessible, usable, reliable, stable, secure and flexible. These features are covered by nonfunctional tests such as performance testing, security testing and load and stress testing.

It's important that nonfunctional testing is measurable – its results must not be characterised by subjective terms such as good or okay.

Some of the types of tests that come under the umbrella of nonfunctional testing are shown below.

Test type	Description
Recovery testing	Tests how well an application can recover from hardware failures or software crashes.
Security testing	Tests how well the systems function as expected by considering confidentiality, integrity and availability. This is to ensure there is no unauthorised access or information leaking to external channels.
Load testing	Tests the ability of the system to function under the real-life conditions of a normal workload. It's a type of performance testing.
Stress testing	Tests the stability, reliability and availability of the system under extremely heavy load conditions to ensure it can recover after a failure. It's sometimes called endurance testing.
Performance testing	Covers many test types that provide information about the quality and stability of the system to inform implementation decisions and validate production design. Some of the test types covered by performance testing are: load, stress, speed and response times, scalability, volume and security.

[Skip to main content](#)

Usability testing	Involves end-users using the system or application to test its ease of use, flexibility and how intuitive it is. This helps designers assess whether the system meets its objectives, including user experience design. It involves more detail than basic usability tests at a functional test level. Usability testing may include accessibility if the business requirements specify this.
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Security testing

Security testing is necessary for large programs or projects.

Examples of security testing are:

- ensuring a password is encrypted
- ensuring unauthorised users cannot access systems or applications they do not have permissions for when processing a financial transaction on a website,
- the back button on the browser should not work,
- preventing erroneous transactions
- ensuring web communication is carried out over HTTPS.



☒ Security testing

Non-Assessed

[Skip to main content](#)

When conducting testing, should you use real data or dummy data in your test environment? Explain why.

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Load testing

Load testing is a type of nonfunctional testing and a type of performance test. The following example demonstrates an example of a load test.

The Open Polytechnic website has in-built analytics that show how many people visit the website, what pages are visited the most and what time of day people tend to visit the most.

According to the analytics, there are more visitors to the website each year. This would be a problem if the infrastructure could not handle the expected number of visitors, resulting in slower page loads and response times.

A load test can be done on the web servers to determine what number of simultaneous visitors to the website will diminish the user experience. A testing tool such as JMeter can be used to create a load benchmark to determine the maximum number of simultaneous visitors possible to still achieve acceptable load and response times.

The analytics and load benchmark can be compared to indicate when overload is likely to occur. Extra servers can be put in place to balance the load for expected visitor numbers.
[Skip to main content](#)

Note: Conducting authorised load testing on a web application you don't own can be considered a form of 'denial of service' attack. Before conducting such tests, it is important to obtain permission to do so. That also means you should not conduct load tests on iQualify 😊.



☒ Load testing

Non-Assessed

What is the difference between load testing and stress testing? How do these types of tests differ in the way they are performed?

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Usability and user acceptance testing

Usability testing and user acceptance testing both contribute to the delivery of a quality product. They are sometimes used interchangeably, but they are not the same – each has its own differences.

Usability testing aims to mimic the end-user's behaviour while using a product or application to see if it works efficiently, effectively and intuitively.

Remember, you're not your user. Testing your product with real users is the best option.

User acceptance testing is performed by the client to make sure the product is working properly **and** according to client requirements. It is often included as a part of beta testing (tests before the software is considered ready for general use) and is generally conducted towards the end of a project to confirm completion. Remember, a product may be functioning correctly, but if it's not doing what the user wants or needs it to do, it doesn't pass a user acceptance test.



☒ Usability testing

Non-Assessed

[Skip to main content](#)

In this activity, you will complete a usability test on the [Yale School of Art website](#).

Use this [usability testing template](#) to conduct your usability test.

Complete the following tasks on the Yale School of Art website, and then complete the usability test.

1. You are interested in the graduate programme at Yale School of Art but want more information from a staff member before applying. Can you find contact details for this?
2. You are interested in visiting the school but aren't sure of their address. Can you find the address for the Yale School of Art on their website?
3. You value the arts and want to give a donation to the Yale School of Art. Can you find information on how to give a donation? Can you find a web page where you are able to donate to the school?
4. You have a visually impaired friend who is also interested in the graduate programme at Yale School of Art. Do you think this website is readable for the visually impaired?

☐ I have completed the activity

Submit



☒ Testing

Non-Assessed

[Skip to main content](#)

Questions: **3**

Now complete the following quiz on functional and nonfunctional testing.

Attempts remaining: **Unlimited**

Start



Thoughts on this page?



Share with your peers:

Your recommendations for enhancing the user experience and overall usability of the Yale School of Art website.

B *I*  

Write your comment

 Add file

Cancel

Send

Skip to main content

Jorge Andres Guerrero Giraldo 22 Feb 2026, 12:07 PM

The Yale School of Art website would benefit from clearer information architecture and more consistent navigation. I would add a simple top menu with key tasks (Programs, Admissions, Contact, Visit, Donate) and ensure the same links appear in the header on every page, not only in the footer. A site wide search bar and better page headings would reduce time spent hunting for information. To improve accessibility, increase text contrast, use consistent font sizes, add clear focus states for keyboard users, and ensure images have alt text. Finally, reduce visual clutter on key pages so primary actions are obvious.



Nicholas Jones 17 Dec 2025, 8:18 AM

Getting a designer to fix the overall design and layout to start. Pick complementary colours and a suitable background for the school. Once that is complete, re-evaluation is needed for further improvement to usability.

